PRACTICAL 1 DBMS

```sql
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;
CREATE TABLE books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    isbn VARCHAR(13) NOT NULL UNIQUE,
    published_year INT,
    CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
CREATE TABLE members (
    member_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE
);
```


```sql
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;
CREATE TABLE books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    isbn VARCHAR(13) NOT NULL UNIQUE,
    published_year INT,
    CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
CREATE TABLE members (
    member_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE
);
```

```sql
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;
CREATE TABLE books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    isbn VARCHAR(13) NOT NULL UNIQUE,
    published_year INT,
    CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
CREATE TABLE members (
    member_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE
);
```

```sql
DESCRIBE members;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

```sql
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Output

Action Output

#	Time	Action	Message
66	20:08:02	USE bookflow_db	0 row(s) affected
67	20:08:22	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT N...	0 row(s) affected
68	20:08:35	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(...	0 row(s) affected
69	20:09:47	DESCRIBE books	4 row(s) returned
70	20:09:58	DESCRIBE members	3 row(s) returned
71	20:10:17	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
22 •  INSERT INTO members (full_name, email) VALUES
23    ('Anil Kumar', 'anil.kumar@example.com'),
24    ('Priya Sharma', 'priya.sharma@example.com'),
25    ('Ravi Verma', 'ravi.verma@example.com');
```

72 20:10:37 INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', '... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
26 •  INSERT INTO books (title, isbn, published_year)
27    VALUES ('Fake Copy', '9780061122415', 2000);
```


```
26 •  INSERT INTO books (title, isbn, published_year)
27    VALUES ('Fake Copy', '9780061122415', 2000);
```


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

```
28 •  INSERT INTO books (title, isbn, published_year)
29    VALUES (NULL, '9999999999999', 2010);
```

83 20:11:19 INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000) Error Code: 1062. Duplicate entry '9780061122415'

84 20:11:41 INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010) Error Code: 1048. Column 'title' cannot be null

```
30 •  INSERT INTO books (title, isbn, published_year)
31    VALUES ('Time Traveler', '8888888888888', 2030);
```

85 20:12:19 INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030) Error Code: 3819. Check constraint 'chk_published_year' is violated.

```
32 •  INSERT INTO members (full_name, email)
33    VALUES ('Anil Clone', 'anil.kumar@example.com');
```

86 20:12:46 INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com') Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'